Interpretable Digital Phenotyping of Subclinical Insomnia from Real-World Wearable Daytime HRV and Sleep Diary Data

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Summary: Wearable daytime heart rate variability (HRV) is seldom examined for subclinical mental health, and pipelines are rarely reproducible end-to-end. In 49 healthy adults monitored for four weeks (public dataset), we derived individualized deviation features from daytime HRV, activity, and sleep diaries, evaluated with Random Forest, SHAP, leave-one-subject-out cross-validation, and 1,000-iteration permutation testing. Insomnia severity was predicted out of sample ($R^2 = 0.068$, $p = 0.004$; HRV complete cases $R^2 = 0.142$, $p = 0.001$), whereas depression, anxiety, and a distress composite were not. Attributions were dominated by self-reported sleep rather than daytime HRV, which bounds the wearable-specific contribution.

Introduction: Consumer wearables permit continuous, real-world monitoring of signals relevant to mental health, yet two limitations occur. Most wearable studies analyse nighttime or resting HRV, leaving daytime HRV underexplored because it is confounded by posture, activity, and time of day. Separately, researchers often prioritize predictive accuracy over interpretability, so it remains unclear how much subclinical signal is recoverable and how much is specific to the wearable rather than co-collected self-report. Detecting variation below diagnostic thresholds is the regime most relevant to prevention and the least studied. Our aim was methodological: to specify and openly evaluate an interpretable, individualized pipeline for subclinical insomnia, test how far any pattern is shared with depression and anxiety, and report the size of the signal.

Methods: We used the publicly available dataset of Baigutanova et al. (2025; CC-BY 4.0): 49 healthy adults (age 28.3 ± 5.9; 25 (51%) female) monitored for four weeks with Samsung Galaxy Watch Active 2 devices, with ISI, PHQ-9, and GAD-7 at three time points. The device was removed overnight for charging, so all HRV is daytime and while awake, and all sleep measures were self-reported. Analyses were restricted to the 08:00–23:00 local (KST) daytime recording window; five-minute HRV metrics (SDNN, LF/HF, heart rate) were residualized on concurrent step count and hour of day, aggregated to daily medians, and combined with sleep-diary measures (latency, wake after sleep onset [WASO], efficiency). For each follow-up assessment, features were deviations from that participant's own rolling baseline (14-day primary, 7-day sensitivity), giving 98 observations from 49 participants; the baseline questionnaire has no preceding data and was excluded. Seven deviation features entered a Random Forest evaluated by leave-one-subject-out cross-validation (LOSO-CV), with significance from 1,000 label permutations, rank stability from 1,000 subject-level bootstraps, TreeSHAP attributions, and a linear mixed-effects model with subject random intercept; grand-mean and subject-mean predictors served as benchmarks. Observations without usable HRV were zero-imputed, with a pre-specified complete-case sensitivity analysis. The identical pipeline was applied to PHQ-9, GAD-7, and a distress composite.

Results: The primary insomnia model generalized to held-out participants: LOSO-CV R^2 0.068 (RMSE 4.39 versus 4.55 for a grand-mean predictor; ISI SD 4.57), permutation $p = 0.004$. Excluding the 14 of 98 zero-imputed observations strengthened the result (84 observations, 45 participants: $R^2 = 0.142$, RMSE 3.99, $p = 0.001$), so the finding is not a missing-data artefact. The effect is modest (~7% of variance); a subject-mean predictor reached RMSE 1.29, confirming between-person level as the dominant source of explainable variance. A mixed-effects model detected no individual gradient (0 of 7 standardized fixed effects at $p < .05$; smallest $p = 0.12$), indicating nonlinear or interactive structure. Attributions were led by WASO and sleep-efficiency deviations, with daytime HRV last; across bootstraps, the three sleep-diary features held the best mean ranks, though rank intervals were wide (mean rank 2.1, interval 1–6 of 7). Secondary outcomes were null (PHQ-9 $R^2 = -0.115$, $p = 0.322$; GAD-7 $R^2 = -0.250$, $p = 0.882$; distress composite $R^2 = -0.095$, $p = 0.254$) despite heavy co-occurrence (observations at cutoff: ISI 46.9%, PHQ-9 36.1%, GAD-7 23.1%; questionnaire $r = 0.59$–0.75; 26.5% of participants exceeded all three), so the signal tracked sleep disruption rather than general distress. The 14-day baseline clearly outperformed the 7-day (R^2 0.068 vs −0.111; $p = 0.004$ vs 0.384).

Conclusion: Individualized deviation features predicted subclinical insomnia severity in held-out participants, and the association survived permutation testing, exclusion of imputed observations, and a change of baseline window. Two qualifications bound this. First, the effect is modest, and stable between-person differences still dominate the outcome in a leave-one-subject-out design. Second, attributions were led by self-reported sleep while daytime HRV ranked last: because the ISI is itself a self-reported sleep instrument, much of the predictive signal plausibly reflects shared method and construct variance. These results should not be read as evidence that wearable daytime HRV predicts insomnia; the specificity to insomnia is consistent with that reading. The contribution is a reproducible, fully specified pipeline with null-model benchmarks, evidence that the 14-day baseline is the more informative specification, and a quantified ceiling on what daytime HRV adds beyond a sleep diary. Generalization is limited by the healthy, narrow cohort and two assessments per participant. Denser designs with ecological momentary assessment, overnight wear providing nocturnal HRV and device-derived sleep, and models that separate sleep-diary from wearable contributions are the necessary next step.